

Zero-Shot Guided Fusion for Class-Incremental Domain Generalization

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Abstract—While Class-incremental learning(CIL) enables continuous category acquisition, existing research predominantly assumes identical training and test distributions, ignoring the critical need for domain generalization(DG) in practical deployments. To this end, We proposed the Class-incremental Domain Generalization(CIDG) benchmark, which is used to test the unseen domain generalization performance of the CIL model trained on multi-source domains. In our experiments, We found that several representative CIL methods actively adjust the parameters based on the source domain, but inadvertently undermine the cross-domain prior knowledge embedded in the pre-trained model(PTM), resulting in its generalization performance often being inferior to the CLIP zero-shot baseline. To ensure that the CIL model can simultaneously meet the DG capabilities provided by PTM and the high performance for specific tasks, we proposed the Zero-Shot Guided Fusion(ZGF). This approach fuses the predictions from the incremental classifier and the CLIP zero-shot classifier after normalization, balancing their contributions through a tunable fusion coefficient. Experiments on CIDG and standard CIL task demonstrate that ZGF substantially improves accuracy on unseen domains while maintaining acceptable performance on standard incremental learning tasks.

Index Terms—Class-incremental learning, domain generalization, pre-trained model, zero-shot learning

I. INTRODUCTION

ASSUMPTIONS in traditional machine learning rely on a “closed-world” scenario where test classes and distributions mirror the training data. However, the real world is dynamic and systems face both domain shifts and unknown categories [1], [2]. These challenges correspond to two critical but isolated research fields: class-incremental learning (CIL) and domain generalization (DG). Despite the flourishing literature on CIL and DG, their interaction remains largely unexplored.

Over the past few years, many CIL methods have been proposed, focusing on how to alleviate catastrophic forgetting and maintain strong performance on all seen classes [3], [4]. Generalizing from these CIL foundations, Continual Learning has found extensive applications in diverse fields [5], [6], [7], [8], [9]. However, almost all existing evaluations are conducted under a single source domain, implicitly assuming that the training and test data are drawn from the same distribution. Similarly, DG methods usually assume static

label spaces and do not account for the continual expansion of categories [10]. In realistic applications, the deployment environment often exhibits substantial distribution shift with respect to the training domains, making it critical to assess not only incremental performance, but also domain generalization ability under continual learning.

Recent research on integrating Incremental Novel Class Discovery and Out-Of-Distribution Generalization has mainly been developed based on two different forms of tasks. The first line extends Domain Adaptation by incorporating unseen class recognition, where tasks defined in [11], [12], [13] assume the availability of target domain data (labeled or unlabeled) to facilitate open-set alignment or test-time adaptation. Another line modifies Class-Incremental Learning by injecting cross-domain data, where scenarios constructed in [14], [15], [16] introduce specific domain shifts or geometric variations directly into the training stream to simulate non-stationarity. However, although these efforts collectively promoted the development of continuous learning and proposed architectures capable of adapting to more realistic and non-static environments, the design of these tasks was still not rigorous enough. Most of these cross-research studies either provide samples from the target domain or merely introduce domain bias in the training data, failing to reflect real-world deployments where data comes entirely from unseen domains.

To bridge this gap, we first design a new evaluation protocol, *Class-Incremental Domain Generalization* (CIDG), which explicitly couples class-incremental learning with leave-one-domain-out testing. In the multi-domain dataset, each domain takes turns to serve as the test set. The model was trained on multiple source domains and tested on an unseen domain. Based on CIDG, we conducted a benchmark test on a series of representative incremental learning methods and discovered that the mainstream CIL approach exhibits significant source domain adaptation bias. This excessive optimization targeting the source domain has impaired the general cross-domain representation ability embedded in the pre-trained backbone network.

Observing that current CIL methods degrade below the original pre-trained CLIP zero-shot baseline [17] due to severe source data overfitting, We considered retaining the original zero-shot capability of the pre-trained model. Inspired by ZSCL [18] and MoE-Adapters [19] which preserve the zero-shot capability of CLIP to identify the out-of-distribution data, we propose a plug-and-play module, Zero-Shot Guided Fusion (ZGF), that directly addresses the more stringent issue of domain generalization. ZGF linearly fuses normalized logits to recover this lost generalization, without modification to the

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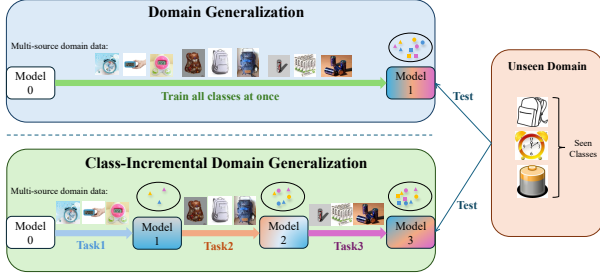


Fig. 1. Unlike DG, which trains once on multi-source data with all classes, CIDG learns new classes task by task on multiple source domains and is evaluated on an unseen domain over all classes seen after the final task.

training pipeline. We conduct extensive experiments on multiple class-incremental tasks and domain generalization tasks, and show that the modified framework significantly improves the DG performance of the model on unseen domains, while keeping the original incremental accuracy sacrifice within an acceptable range. The main contributions of this work include:

- We propose a new evaluation task, Class-Incremental Domain Generalization, to systematically assess the domain-generalization capability of CIL methods.
- We conducted extensive experiments on several representative CIL methods to measure their domain generalization performance within the CIDG framework.
- We present Zero-Shot Guided Fusion, a simple yet effective framework that linearly fuses normalized logits from the incremental classifier and a fixed CLIP zero-shot discriminator, yielding substantial improvements on unseen domains while incurring only negligible loss in class-incremental performance.

II. METHODOLOGY

A. Class-Incremental Domain Generalization

We adopt the DomainBed [20] evaluation framework for domain generalization, integrating the special training procedure of class-incremental learning to design the Class-Incremental Domain Generalization benchmark. As shown in Fig. 1, models undergo class-incremental training across multiple source domains, with new categories introduced in each task; after each training round, performance is evaluated on an unseen target domain using all classes seen so far. The dataset that is compatible with CIDG must contain multiple domains, and each domain will be successively used as a test set for unseen domains. Evaluation metrics include Last-Task Accuracy (A_B) and Average Accuracy ($\bar{A}$), measuring final model performance and stability across the entire learning trajectory, respectively.

Let $\mathcal{X}$ be the input space and $\mathcal{Y}$ the label set. Given M source domains $\mathcal{D}_S = \{\mathcal{D}^{(s)}\}_{s=1}^M$, each characterized by joint distribution $P^{(s)}(x, y)$. A target domain $\mathcal{D}^{(\tau)}$ is reserved for testing only. Class-incremental learning comprises T tasks, where task t introduces disjoint class subset $\mathcal{Y}_t$, with cumulative label set $\mathcal{Y}_{1:t} = \bigcup_{i=1}^t \mathcal{Y}_i$. Model f_{θ_t} outputs logits over $\mathcal{Y}_{1:t}$ after task t , with prediction

$$\hat{y}_t(x) = \arg \max_{y \in \mathcal{Y}_{1:t}} f_{\theta_t}(x)_y. \quad (1)$$

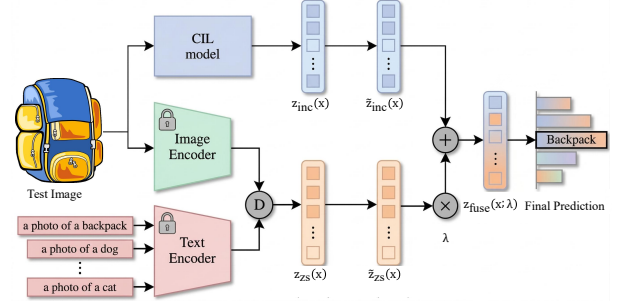


Fig. 2. Architecture of Zero-Shot Guided Fusion. On the top is the branch of the CIL model trained on multiple source domains, and below is the branch of CLIP for zero-shot learning. The incremental classifier and CLIP zero-shot discriminator provide complementary logits, which are L2-normalized and fused with coefficient λ .

CIDG differs from conventional CIL in that testing occurs on the unseen target domain $\mathcal{T}_t^{(\tau)}$, with task- t risk defined as

$$\mathcal{R}_t^{(\tau)}(f_{\theta_t}) = \mathbb{E}_{(x,y) \sim P^{(\tau)}} [\ell(\hat{y}_t(x), y)]. \quad (2)$$

CIDG is essentially a domain generalization problem with a specific training procedure. Fixing the CIL algorithm $\mathcal{A}$, the entire multi-task training can be viewed as a complex transformation producing hypothesis f_{θ_T} from multi-source data. The outer-level objective minimizes the target domain expected risk $\mathcal{R}_{\text{CIDG}}^{(\tau)}(\mathcal{A})$, subject to the constraint that θ_T is obtained via incremental updates. The core goals of CIDG and DG are the same in general, but there are differences in the training process. Thus, CIDG can be viewed as a natural extension of standard DG to the incremental setting, preserving the theoretical validity of evaluating out-of-distribution generalization.

B. Zero-Shot Guided Fusion

The CIDG benchmark reveals that most CIL methods, although strong on the on data without domain shift, may still suffer noticeable degradation on the unseen domain. At the same time, a frozen CLIP model provides a zero-shot discriminator for all labels in $\mathcal{Y}$ that is often robust to domain shift but ignores incremental supervision. Zero-Shot Guided Fusion (ZGF) is designed to combine these two complementary decision sources.

A direct gating strategy that selects either the incremental model or CLIP based on confidence is generally ineffective in CIDG. Although CLIP exhibits strong cross-domain priors, its predictions are often less confident than those of the incremental classifier [21], especially on categories that have been extensively trained. As a result, gating strategies based on simple confidence tend to favor incremental models almost everywhere and fail to take full advantage of zero-shot cues. As Fig. 2, ZGF circumvents this issue by fusing normalized logits instead of hard switching between models. MoE-Adapters [19] and CICA [22] demonstrated the effectiveness of multi-expert collaboration, supporting our fusion strategy's feasibility. Unlike MoE-Adapters which use CLIP for zero-shot defense against new category intrusions, ZGF focuses

on directly enhancing the model’s cross-domain generalization ability and is a plug-and-play module that requires no additional training.

Let f_{θ_T} be a trained incremental classifier as above, and let g_{CLIP} denote a frozen CLIP model with a set of text prompts $\{t_y\}_{y \in \mathcal{Y}}$. For an input x , we define:

- **Incremental logits:**

$$z_{\text{inc}}(x) \in \mathbb{R}^{|\mathcal{Y}_{1:T}|}, z_{\text{inc}}(x)_y = f_{\theta_T}(x)_y, \quad (3)$$

restricted to classes seen so far.

- **Zero-shot logits:**

$$z_{\text{zs}}(x) \in \mathbb{R}^{|\mathcal{Y}_{1:T}|}, z_{\text{zs}}(x)_y = \alpha \langle \phi_{\text{img}}(x), \phi_{\text{text}}(t_y) \rangle, \quad (4)$$

where ϕ_{img} and ϕ_{text} are CLIP image and text encoders, and α is the CLIP logit scale.

Since the two logit vectors may have different scales and calibration, ZGF first applies per-sample L_2 normalization:

$$\tilde{z}_{\text{inc}}(x) = \frac{z_{\text{inc}}(x)}{\|z_{\text{inc}}(x)\|_2 + \varepsilon}, \tilde{z}_{\text{zs}}(x) = \frac{z_{\text{zs}}(x)}{\|z_{\text{zs}}(x)\|_2 + \varepsilon}, \quad (5)$$

with a small constant $\varepsilon > 0$ for numerical stability. The fused score is then defined as

$$z_{\text{fuse}}(x; \lambda) = \tilde{z}_{\text{inc}}(x) + \lambda \tilde{z}_{\text{zs}}(x), \quad (6)$$

where $\lambda \geq 0$ is a scalar fusion coefficient. The final ZGF prediction is

$$\hat{y}^{\text{ZGF}}(x; \lambda) = \arg \max_{y \in \mathcal{Y}_{1:T}} z_{\text{fuse}}(x; \lambda)_y. \quad (7)$$

On the whole, the incremental branch encodes task-specific knowledge learned from the source domains, while the zero-shot CLIP branch supplies cross-domain semantic priors. The normalization step removes scale differences and forces the two branches to contribute through their *direction* in logit space. This normalization eliminates scale discrepancies, making the fusion coefficient λ consistent across diverse domains and less sensitive to dataset-specific variations. By tuning the λ , ZGF smoothly fuse the pure CIL prediction and CLIP-dominated zero-shot prediction.

III. EXPERIMENTS

We conduct a systematic benchmark experimental evaluation of a variety of representative class incremental learning methods under the framework of the CIDG benchmark in Section II. All implementations are based on the open-source class-incremental toolbox C3Box [23], upon which we build our CIDG evaluation pipeline.

A. CIDG Benchmark

We instantiate CIDG on three widely used domain generalization benchmarks: Terra Incognita [24], OfficeHome [25] and DomainNet [26]. To keep the incremental difficulty comparable across datasets with very different label cardinalities, we fix the number of incremental steps to ($T = 5$) and partition the full label set into five disjoint subsets of approximately equal size. We compared a wide range of representative CIL methods based on the assessment criteria of CIDG, and all

TABLE I
PROPOSED CIDG BENCHMARK RESULTS.
 A_B : LAST-TASK ACCURACY. $\bar{A}$: AVERAGE ACCURACY OVER TASKS.

Category	Methods	Terra Incognita [24]				OfficeHome [25]				DomainNet [26]				Average			
		A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)
Baseline	zs_CLIP [17]	30.18	55.21	86.05	87.82	59.61	65.27	58.61	69.43								
Data Replay	MEMO [28]	25.46	43.95	22.58	47.18	16.57	38.67	21.54	43.27								
	FOSTER [29]	24.84	45.51	49.52	67.99	30.47	36.64	34.94	50.05								
Backbone Expansion	EASE [30]	28.01	53.87	84.24	87.67	48.34	50.54	53.53	64.03								
Prompt Expansion	L2P [31]	23.63	43.87	21.55	42.22	15.14	37.29	20.11	41.13								
	MEMO [28]	25.46	43.95	22.58	47.18	16.57	38.67	21.54	43.27								
	FOSTER [29]	24.84	45.51	49.52	67.99	30.47	36.64	34.94	50.05								
Knowledge Distillation	PROOF [34]	28.58	56.43	84.42	88.10	46.48	52.79	53.16	65.77								
	PROOF [34]	28.58	56.43	84.42	88.10	46.48	52.79	53.16	65.77								
Model Rectify	RAPF [35]	43.57	60.98	85.42	88.46	47.41	51.35	58.80	66.93								
Prototype/Template-Based Classification	SimpleCIL [16]	26.98	56.91	84.77	88.33	50.65	55.78	54.13	67.01								
	Aper_adapter [16]	29.52	54.59	84.09	87.51	50.10	56.49	54.57	66.20								
	Aper_ssf [16]	30.30	54.73	83.45	86.88	49.58	55.98	54.44	65.86								
	Aper_vpt_deep [16]	30.04	56.04	83.88	87.34	46.60	52.99	53.51	65.46								
	Aper_vpt_shallow [16]	28.46	54.69	84.06	87.48	50.12	56.61	54.21	66.26								
	Aper_finetune [16]	28.38	54.46	83.17	87.15	45.14	52.25	52.23	64.62								
	TUNA [36]	40.52	60.29	86.19	89.25	54.90	59.15	60.54	69.56								
	CLG-CBM [37]	42.35	55.77	85.03	87.17	49.07	55.48	58.82	66.14								

were uniformly constructed using the CLIP ViT-B/16 [27] backbone network. The baseline we choose the CLIP zero-shot [17].

Table I shows that Data Replay methods (43.27–50.05) and Prompt Expansion methods (41.13–48.51) perform poorly in CIDG, falling 19–26 points below CLIP zero-shot [17] at 69.43. These approaches modify the pretrained model too aggressively, destroying cross-domain generalization [38]. Backbone Expansion via EASE [30] at 64.03 and Knowledge Distillation via PROOF [34] at 65.77 fare slightly better but still fall short of the baseline. Model Rectify via RAPF [35] at 66.93 is the first to exceed zero-shot performance, suggesting that mild regularization helps maintain the pre-trained structure. Among Prototype/Template-Based methods, SimpleCIL [16] and Aper [16] variants range from 64.62 to 67.01 and underperform as they only use the image encoder without text branch semantics. TUNA [36] at 69.56 and CLG-CBM [37] at 66.14 achieve better results by applying discriminative adaptation to the full CLIP architecture. Overall, methods with lighter constraints on the pretrained model preserve cross-domain generalization better than those with heavy structural modifications.

B. ZGF under CIDG

It is important to note that not all CIL methods are suitable for using ZGF. The SimpleCIL and Aper [16] variants rely entirely on prototype-based nearest neighbor classification in the feature space and do not have explicit parametric classifier headers, making it important to interface them with the logit-level fusion used in ZGF. On the other hand, methods such as RAPF [35], TUNA [36], and CLG-CBM [37] have achieved CIDG performance above the CLIP zero-shot [17] baseline, and further increasing the impact of CLIP may only yield marginal DG gains at the cost of a noticeable reduction in traditional CIL metrics, which is limited in practical terms.

As shown in Fig. 3, we set a range for λ and then adjusted the parameters within that range. We observe a consistent pattern across datasets when we adjust the parameters: as λ grows from zero to a moderate range, CIDG performance on unseen domains improves substantially, often surpassing the CLIP zero-shot reference. When λ becomes too large, the fused model approaches pure zero-shot prediction and the gains

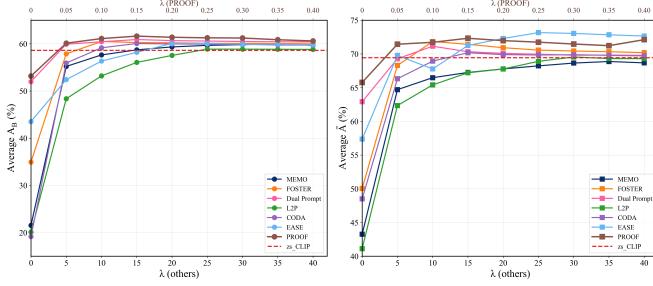


Fig. 3. The average results of parameter tuning on the three datasets mentioned in Section III-A.

TABLE II
CIDG RESULTS WITH ZGF UNDER VARYING FUSION COEFFICIENT λ .

Base Methods	ZGF(Ours)	λ	Terra Incognita [24]		OfficeHome [25]		DomainNet [26]	
			A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)
zs_CLIP [17]	—	—	30.18	55.21	86.05	87.82	59.61	65.27
MEMO [28]	✓	35	30.83 ^{+5.37}	50.80 ^{+6.85}	89.17 ^{+66.59}	90.18 ^{+43.00}	59.85 ^{+43.28}	65.64 ^{+26.97}
FOSTER [29]	✓	10	31.20 ^{+6.36}	57.62 ^{+12.11}	90.07 ^{+40.55}	91.30 ^{+23.31}	60.21 ^{+29.74}	66.58 ^{+29.94}
Dual Prompt [32]	✓	25	30.86 ^{+7.23}	50.02 ^{+6.15}	85.96 ^{+64.41}	89.36 ^{+47.14}	59.79 ^{+44.65}	66.31 ^{+29.02}
L2P [31]	✓	15	32.08 ^{+4.00}	53.70 ^{+9.33}	89.83 ^{+6.89}	90.69 ^{+3.72}	60.80 ^{+15.84}	66.58 ^{+9.18}
CODA [33]	✓	10	30.45 ^{+11.91}	52.60 ^{+5.19}	89.79 ^{+67.28}	90.89 ^{+30.36}	57.23 ^{+40.81}	63.31 ^{+25.72}
EASE [30]	✓	20	31.26 ^{+3.25}	63.40 ^{+9.53}	88.44 ^{+4.20}	89.48 ^{+1.81}	60.53 ^{+42.19}	63.97 ^{+33.43}
PROOF [34]	✓	0.15	34.14 ^{+5.56}	59.43 ^{+3.00}	89.93 ^{+5.51}	90.95 ^{+2.85}	60.80 ^{+14.32}	66.58 ^{+13.79}

Note: The red superscripts denote the performance gain compared to the baseline results without ZGF in Table I.

saturate or slightly decline. Based on this phenomenon and concerns about excessive correction amplitude, we selected set of optimal λ .

Table II reports the CIDG results of several representative CIL methods under the optimal fusion coefficient λ obtained by our parameter adjustment. The selected models all bridge the performance gap with the CLIP zero-shot baseline in the CIDG setting, with the methods with poor base generalization ability benefiting the most. Some of the methods even outperform the performance of zs_CLIP, which indicates that ZGF effectively utilizes the complementarity between incremental supervision and zero-shot semantics.

C. ZGF under Standard CIL Benchmarks

A desirable fusion scheme should significantly boost unseen domain performance in CIDG, while causing only minor degradation on standard source domain CIL benchmarks. To verify whether ZGF is equally practical beyond CIDG, we further evaluated it on the traditional identically distributed class-increment benchmark. The four selected datasets include ImageNet-R [39], CIFAR-100 [40], CUB-200 [41], and Food-101 [42], covering image classification tasks of different scales and degrees of fineness.

For each model mentioned in Section III-B, we directly reuse the fusion coefficient λ we selected in Table II, without re-tuning the parameters in the original CIL experiment. Such a design can test whether the λ chosen in the domain generalization scenario still gives an acceptable compromise on the identically distributed CIL benchmark, thus testing the practical value of ZGF.

Table III shows that ZGF maintains stable performance on standard CIL benchmarks, even yielding slight improvements on ImageNet-R [39] and Food-101 [42] due to the regularization effect of the zero-shot branch, while the performance on

TABLE III
ZGF PERFORMANCE ON STANDARD CIL BENCHMARKS.

Base Methods	ZGF(Ours)	ImageNet-R [39]		CIFAR100 [40]		CUB200 [41]		Food101 [42]		Average	
		A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)	A_B (%)	$\bar{A}$ (%)
MEMO [28]	✓	75.80	85.32	75.15	85.95	79.01	87.34	75.86	86.38	76.46	86.25
	✓	78.73	85.38	74.16	83.77	77.10	84.32	83.41	89.16	78.35	85.66
FOSTER [29]	✓	81.15	86.46	81.99	88.30	81.55	84.85	82.71	88.80	81.85	87.10
	✓	82.98	87.95	78.88	86.60	79.43	84.93	84.94	90.25	81.56	87.43
Dual Prompt [32]	✓	74.60	81.00	76.80	83.23	77.40	85.14	74.64	82.13	75.86	82.88
	✓	78.13	84.83	74.27	83.84	69.04	79.77	82.96	88.75	76.10	84.30
L2P [31]	✓	77.10	83.50	75.41	83.97	76.00	85.06	74.59	82.79	75.78	83.83
	✓	78.12	84.71	73.27	83.31	69.72	80.40	82.88	88.64	76.00	84.27
CODA [33]	✓	73.92	83.94	78.30	86.84	74.94	84.75	75.75	85.46	75.73	85.25
	✓	78.55	85.48	76.26	85.60	72.86	82.73	83.45	89.38	77.78	85.80
EASE [30]	✓	73.45	80.55	74.28	82.95	74.47	82.93	72.72	80.18	73.73	81.65
	✓	77.50	84.16	71.69	82.01	64.08	75.76	82.12	87.99	73.85	82.48
PROOF [34]	✓	78.20	83.98	78.83	86.26	78.50	85.37	80.36	87.73	78.97	85.84
	✓	78.93	84.91	79.06	86.48	78.88	85.51	80.93	88.10	79.45	86.25

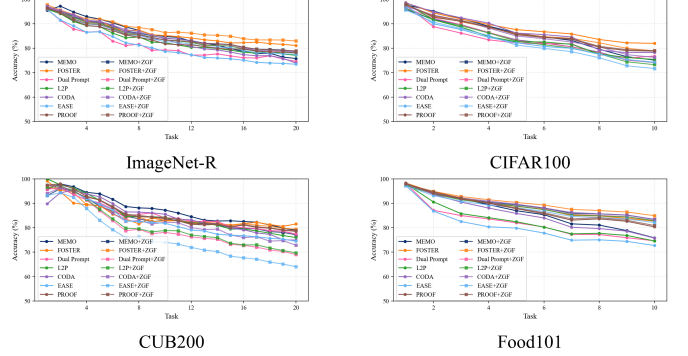


Fig. 4. Per-task accuracy on standard class-incremental learning with a base-0 setup and 10 new classes per task, comparing representative CIL methods with and without ZGF. The dots represent the original CIL baseline, and the square dots represent the ZGF superimposed during the inference stage.

CIFAR-100 [40] remains comparable. The drawback is that a noticeable drop occurs on the fine-grained CUB200 [41] dataset, where the coarse-grained semantic priors of CLIP may interfere with the local features required for fine discrimination. Nevertheless, the average results remain basically stable overall, making this minor trade-off acceptable given the substantial generalization gains in the CIDG scenario.

IV. CONCLUSION

This letter reveals for the first time the critical blind spot of domain generalization in class-incremental learning, establishes the CIDG benchmark, and systematically evaluates representative CIL methods. We further propose a plug-and-play ZGF framework that significantly enhances generalization to unseen domains while maintaining competitive performance on standard CIL tasks.

Future work. ZGF is essentially a multi-model fusion strategy that uses pre-trained CLIP branches to guide the decision in the inference phase, which inevitably incurs additional computational overhead for real-time systems. Therefore, an important direction for future class incremental learning is to jointly optimize the objectives of CIL and DG tasks in the model training phase, and better internalize the zero-shot semantic prior of the pre-trained model into a single model, ensuring that a single model is sufficient to maintain excellent performance on both old and new classes on the source domain and the unseen domain.

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